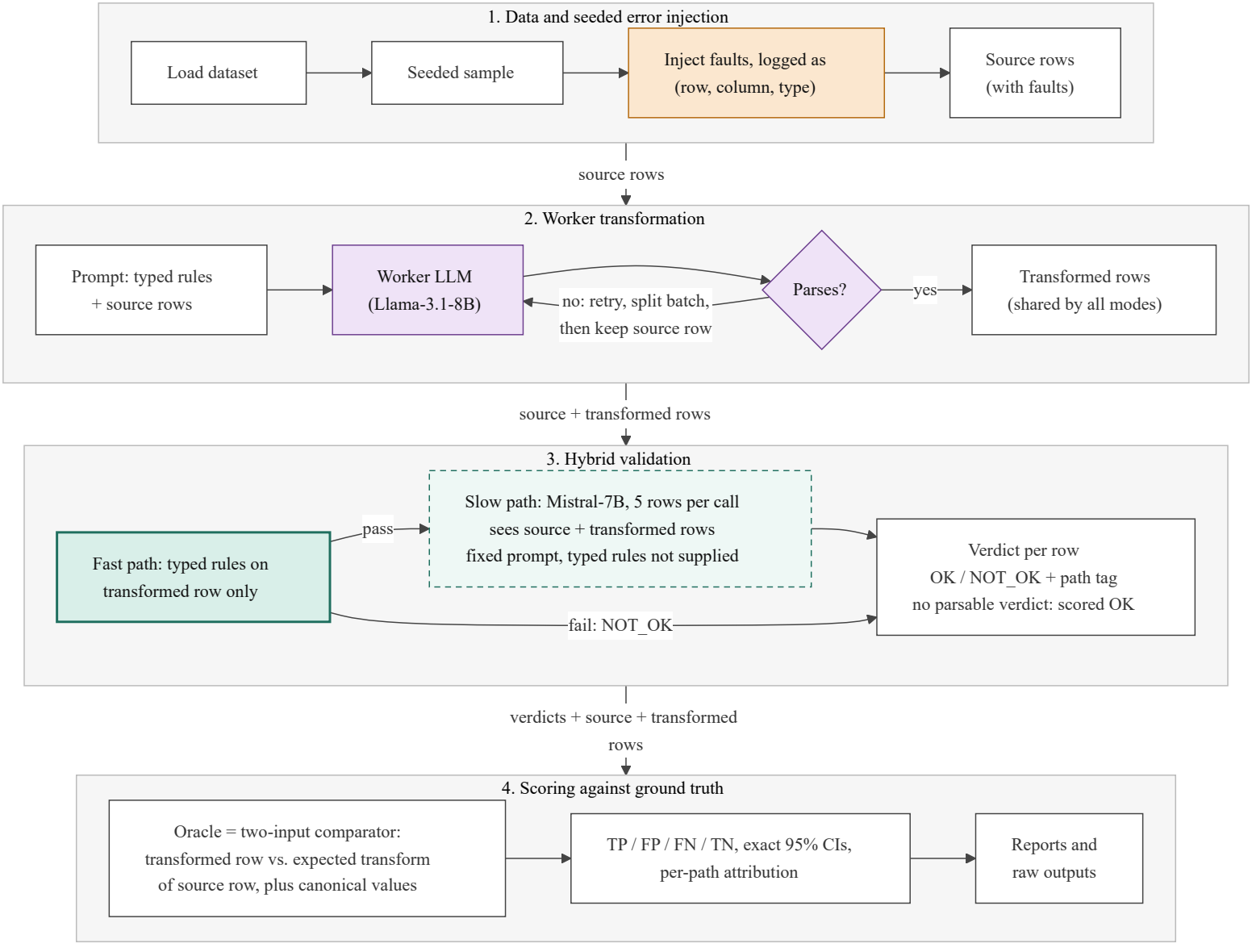

1. Data and seeded error injection
Load dataset
Seeded sample
Inject faults, logged as
(row, column, type)
Source rows
(with faults)
source rows
2. Worker transformation
Prompt: typed rules
+ source rows
Worker LLM
(Llama-3.1-8B)
Parses?
yes
Transformed rows
(shared by all modes)
no: retry, split batch,
then keep source row
source + transformed rows
3. Hybrid validation
Slow path: Mistral-7B, 5 rows per call
sees source + transformed rows
fixed prompt, typed rules not supplied
Fast path: typed rules on
transformed row only
pass
fail: NOT_OK
Verdict per row
OK / NOT_OK + path tag
no parsable verdict: scored OK
verdicts + source + transformed
rows
4. Scoring against ground truth
Oracle = two-input comparator:
transformed row vs. expected transform
of source row, plus canonical values
TP / FP / FN / TN, exact 95% CIs,
per-path attribution
Reports and
raw outputs